

A Depth-Aware, Physics-Based Data Augmentation Framework for Simulating Rain and Fog Distortions in Outdoor-RGB Images

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Abstract

1 Introduction

Images used in computer vision are often distorted. Distortion in computer vision refers to any signal that produces visual deviation in image/video from its original expectation. When the original image/video is unavailable, an observed visual comfort or simply annoying effect on the aesthetic visual appearance is also considered as a distortion [1]. These distortions could be sorted as In-Capture Distortion, including blur, noise, uneven illumination, color shifts, haze, rain, snow, and other visibility changes; and Post-Capture Distortion, such as transmission, impairments, enhancement artefacts, and adversarial distortion. Although these effects may appear minor to human observers, they can change the statistical structure of an image and reduce the reliability of machine learning models trained primarily on clean data [1]. Distortions affect the performances of high-level tasks, such as object classification, object detection, and scene understanding [1]. Therefore, distortion detection and removal are a possible preprocessing step for high-level tasks to decrease the impact on performance.

For machine learning, distortion is not only an image-quality issue but also is affected by dataset and training. Models usually learn from the distribution of inputs they are given, so a model trained only on clean images may have poor generalizability when working on the same object appearing under poor visibility, weather effects, or degraded contrast. Distortion-specified datasets are important because they make it possible to evaluate model robustness by type and severity, rather than reporting only a single clean-data accuracy score [1].

Current datasets of natural distortions have important limitations. Real rain, haze, fog, and snow images can be difficult to collect at scale, and the resulting data may be unbalanced across weather type, severity, scene category, and annotation quality. Many benchmark datasets also include distortions as broad corruptions rather than as physically

meaningful, scene-dependent effects. This limits how well they can support controlled robustness studies or targeted training strategies. The need for environment-aware image formation is also visible in underwater computer vision, where more complicated distortions present due to an uneven refractive index of water, such as attenuation, scattering, color-shifting, halo effect, etc. [2].

Related work has explored both distortion recognition and natural-distortion simulation. Survey work on distortion classification emphasizes that multiple and natural distortions remain difficult to categorize and model consistently [1]. In weather-focused vision, rain simulation has been used to render controlled rain effects for robustness evaluation, while haze and fog modeling often relies on atmospheric image-formation ideas such as scattering, attenuation, and reduced scene contrast. Work on adverse-weather perception, including foggy driving scenes, also shows that real weather can strongly affect downstream recognition and detection performance.

The model trains a distortion augmentation framework, which adds controlled rain and fog effects to existing baseline datasets such as BDD100K, ACDC and Cityscapes [3, 4, 5]. The framework is evaluated by comparing the performances of a downstream object detection model, YOLOv8n [6, 7], on the original dataset and the augmented dataset.

1.1 Cycle-GAN (Generative Adversarial Network)

Generative Adversarial Network (GAN) is a deep learning model that consists of two networks that compete against each other: a generator G and a discriminator D . The generator G tries to generate realistic images from a target domain X , while the discriminator D tries to discriminate generated images from real images. That is, the generator learns a mapping: $G : X \rightarrow Y$. For example, in the context of our

work, the generator G tries to create rainy or foggy looking images from clear images, while the discriminator D tries to distinguish between real rainy or foggy images and generated images. The generator and discriminator are trained in an adversarial manner, where the adversarial framework’s objective is to minimize the generator loss and maximize the discriminator loss. The loss function of GAN is given below, defined by Goodfellow et al[8].

$$\min_G \max_D \mathbb{E}_{x \sim p_{\text{data}}(x)} [\log D(x)] + \mathbb{E}_{z \sim p_z(z)} [\log(1 - D(G(z)))]. \quad (1)$$

Traditional GANs use paired training data, which consists of training examples $\{x_i, y_i\}_{i=1}^N$, where x_i is the input and y_i is the corresponding output. GAN is useful for image-to-image translation tasks, such as photo reconstruction from edge maps, colorization, photo synthesis from label maps, etc[9]. However, in many cases, paired training data is not available. In the context of our work, it is difficult to collect paired training data of clear images and corresponding rainy or foggy images at same places. Existing datasets, such as BDD100K, cityscapes, or ACDC, are all unpaired datasets[3, 5, 4]. Figure 1 illustrates the difference between paired and unpaired training data.

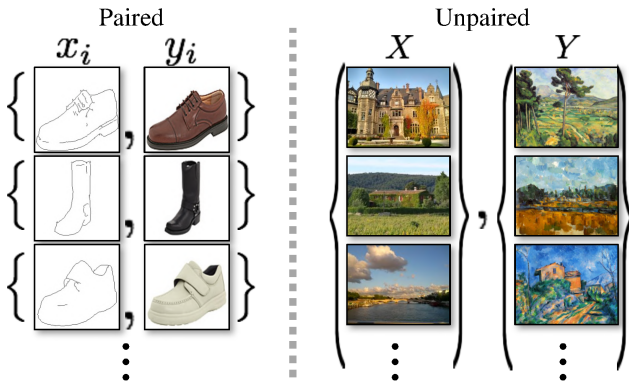


Figure 1: *Paired* training data (left) consists of training examples $\{x_i, y_i\}_{i=1}^N$. *Unpaired* training data (right) consists of a source set $\{x_i\}_{i=1}^N$ ($x_i \in X$) and a target set $\{y_i\}_{i=1}^N$ ($y_i \in Y$). Unpaired training contains no correspondence from the source set X to the target set Y . Figure adapted from [10].

Our work uses Cycle-GAN, which is a type of GAN that learns from unpaired training data. Cycle-GAN uses two generators and two discriminators. A generator $G : X \rightarrow Y$ transforms images from domain X to domain Y ; a reverse generator $F : Y \rightarrow X$ transforms images from domain Y to domain X . The first discriminator D_Y distinguishes between real images in domain Y and generated images from

generator G ; the second discriminator D_X distinguishes between real images in domain X and generated images from generator F . The Cycle-GAN framework uses a cycle-consistency loss to ensure that the generated images can be transformed back to the original domain. The cycle-consistency loss is given below, defined by Zhu et al[10].

$$\mathcal{L}_{\text{cyc}}(G, F) = \mathbb{E}_{x \sim p_{\text{data}}(x)} [\|F(G(x)) - x\|_1] + \mathbb{E}_{y \sim p_{\text{data}}(y)} [\|G(F(y)) - y\|_1]. \quad (2)$$

The objective of Cycle-GAN is to minimize the adversarial loss, maximize the discriminator loss, and minimize the cycle-consistency loss. The key idea for Cycle-GAN is that if an image is transformed from domain X to domain Y by G , and then it is transformed back to domain X by F , the resulting image should be similar to the original image. This cycle-consistency allows the model to not only produce an image that looks like it belongs to the target domain, but also preserves the content, or the structure of the original image[10].

In weather augmentation, cycle-GAN is useful because it generates realistic rainy or foggy images from clear images while it preserves the scene structures and it does not require paired training data. However, a limitation of Cycle-GAN, as well as other GANs, is that it is not controllable. It is impossible to control how much wetness, rain, or fog is generated, and it is impossible to control the depth-dependent effects of rain and fog[10, 11]. Therefore, our work uses Cycle-GAN to generate rainy and foggy images, but it also uses a physics-based, depth-aware renderer to generate rain with controllable appearance.

2 Related Work

2.1 Distortion Classification and Evaluation

Traditional distortion classification models and evaluation use blur indices, noise estimators, natural scene statistics, image quality metrics, Support Vector Machines (SVM), and K-Nearest Neighbours (KNN) to evaluate the impact of distortions. Recent distortion classification and detection models use Convolutional Neural Networks (CNN) and temporal models. Ahn, Kang, and Sohn train a Y-shaped CNN for joint distortion classification and localization. Other methods for distortion classification include feature pool, relating metrics with specific distortions, and Natural Scene Statistics (NSS). The method supports learned distortion features, yet most datasets focus on controlled distortions rather than natural weather [1, 12].

The quality of visual content in images and distortion severity are important factors that affect performance in high-level tasks, including object classification, object detection, and scene understanding. Using distortion classification and

detection models could help researchers finetune models that are more resistant to distortions. However, our work helps finetuning models by presenting a data augmentation framework that increases models’ resilience to rain and fog distortions from training, which does not require distortion detection and distortion removal process.

2.2 Current Limitations of Distortion Datasets

Current distortion datasets give labels and severity levels, but many examples come from synthetic, single-type, quality-assessment settings. LIVE, TID2008, TID2013, CSIQ, Waterloo, KADID-10K, LIVEMD, and MDID help keep a controlled evaluation. However, they do not contain natural weather with precise physical labels. Rain, haze, fog, and snow depend on lighting, exposure, particle density, scene depth, and occlusion. There is a lack of labeled, controlled multi-distortion datasets. Consequently, classification and localization of multiple distortions remains challenging due to lack of labeled, controlled datasets [1, 11]. Our work adds controlled, artificial distortions, which addresses lack of labeled, controlled datasets, especially for outdoor datasets.

2.3 Traditional Rain Rendering

Garg and Nayar model rain streak photometry and spatio-temporal behavior. Their work links rain visibility to drop size, velocity, exposure time, scene radiance, camera settings, and drop shape oscillation. This result matters for rendering because a transparent 2D texture ignores camera and scene factors. Traditional rain rendering lacks the effect of moisture on the scene, and it often appears to be unrealistic. For instance, traditional rain rendering lacks wet surfaces such as water on the ground, and it creates raindrops that are supposed to be hidden behind buildings [11]. Though, this framework supported rain detection, removal, rendering, and camera-based rain measurement [13].

2.4 Modern Rain Rendering

Modern rain rendering turns rain models into dataset augmentation for robustness tests. Halder, Lalonde, and de Charette combine a particle simulator, outdoor illumination estimation, and rain photometry to render rain and fog over clear driving images. The pipeline preserves labels from KITTI and Cityscapes while changing weather appearance. Detection and segmentation results then measure model sensitivity under controlled rain [14].

Tremblay, Halder, de Charette, and Lalonde compare three rendering routes: physics-based, data-driven, and hybrid. They classify raindrops into rain particles and fog-like rain. Fog-like rain is a single pixel that contains multiple rain

particles; it is located far away from the camera and appears as fog. The physics-based route combines fog-like rain, rain particles, and photometric modeling. The renderer estimates outdoor illumination because a raindrop receives light from a wider angular range (approximately 160 degrees) than the camera’s view (70 to 100 degrees). This design improves physical control over a flat overlay. The same design also depends on sky visibility, environmental symmetry about photometry, and compute cost [11, 15, 16].

The data-driven route uses a GAN-style image translation model. A GAN trains two networks: a generator synthesizes target-domain images, and a discriminator judges whether each image looks real or generated. Adversarial training on the generator outputs rainy-looking images. The output from the generator contains features such as dark sky, rainy environment, wet surfaces and dim lamination. However, GAN is less controlled, which risks rain density, depth ordering, and scene consistency. GAN’s focus on global appearance also creates unrealistic features, such as fake water reflections [11, 9, 17].

Tremblay et al. render rain over KITTI, Cityscapes, and nuScenes. They evaluate object detection, semantic segmentation, and depth estimation under increasing degradation. The paper reports degrading performances as increasing rain intensity and more robust performances after finetuning using rain-augmented datasets. The paper reports that outputs from GAN only are user-rated as the most realistic images; outputs from GAN plus PBR perform the best on data augmentation [11].

Our work is inspired from the framework from Tremblay et al. our work uses the traditional rain rendering framework with depth estimation to build rain streaks with controlled, depth-dependent rain streaks; our work also involves using a Cycle-GAN model that adds rainy and foggy visual effects.

2.5 Fog Rendering

Fog rendering often follows the atmospheric image formation model. Scene radiance attenuates with distance, while airlight grows along the camera’s ray. Sakaridis, Dai, and Van Gool applies this model to clear Cityscapes images and produce Foggy Cityscapes. Depth makes fog stronger for far regions rather than uniform across the image. Training with synthetic fog improves semantic foggy scene understanding, but results depend on depth quality, density choices, and the synthetic-to-real gap [18].

Clear2Fog extends this direction with monocular depth estimation and atmospheric light estimation. Monocular depth estimation does not require 3D constructions such as LiDAR; rather, this paper uses Apple Depth Pro to produce an estimated depth map. The pipeline simulates fog on clear-weather driving datasets and tries to preserve camera-LiDAR consistency. Without relying on LiDAR, Clear2Fog remains

stable under extreme conditions like fog, snow or haze. Seeing Through Fog Without Seeing Fog focuses on multimodal perception rather than rendering. The dataset and fusion model show different failure patterns across camera, LiDAR, radar, and gated near-infrared sensors [19, 20].

2.6 Depth Estimation and Application in Distortion Rendering

Depth provides the geometry needed for natural distortion placement. In haze and fog, depth sets attenuation and airlight strength. In rain, depth helps determine the severity, intensity and placement of rain streaks. A renderer without geometry often places plausible-looking weather in physically inconsistent locations [18, 11].

Depth estimation models estimate distance from an image or complete sparse sensor depth. The output supplies an approximate geometry map when a dataset lacks dense ground truth. Jaritz et al. study sparse and dense depth completion for driving scenes. Depth Anything V2 offers a modern monocular option trained with synthetic images and pseudo-labeled real images. This makes Depth Anything V2 useful for RGB-only distortion rendering [21, 22].

Our work uses the depth-map output from Depth Anything V2 as the guiding parameter for transparency and appearance of rain streaks and translucency and intensity of fog. The depth-map output from Depth Anything V2 is stable and reliable, and it is a strong model for completing our computer vision tasks.

2.7 Datasets With Distortions

General distortion and corruption datasets support baselines, but natural weather rendering requires more geometry and scene context. The distortion survey lists LIVE, TID2008, TID2013, CSIQ, Waterloo, KADID-10K, LIVEMD, and MDID as major image-quality or multi-distortion resources. ImageNet-C standardizes common corruption robustness for image classifiers. These resources help robustness framing. They do not model depth-aware rain, haze, or fog. [1].

Weather-specific datasets match the project more closely. ACDC includes fog, nighttime, rain, and snow with dense driving-scene annotations and normal-condition correspondences for many scenes. Foggy Cityscapes and rain-augmented KITTI, Cityscapes, and nuScenes preserve existing labels while adding synthetic weather. The limitation of these datasets is the lack of controlled severity or intensity, depth-aware labels, and controlled natural distortions [4, 18, 11].

2.8 BDD100K

BDD100K serves as a baseline dataset for our work, because the data are large, diverse, and driving-focused. The dataset contains 100,000 driving videos and supports detection, segmentation, lane marking, drivable area, and related driving labels. BDD100K also includes geographic, environmental, and weather variation. The limitation is missing paired clean/adverse views and controlled severity labels. A depth-aware rain and haze renderer would turn BDD100K into a controlled robustness testbed while keeping real driving diversity. [23].

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